

# 120 Days of Quant Finance

Week 2 – Statistical Inference

Day 1: Central Limit Theorem

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## Introduction

One of the most fundamental results in probability theory and statistics is the Central Limit Theorem (CLT). It explains why Normal distributions appear so frequently in statistical modeling and inference, even when the underlying data is not Normally distributed.

The Central Limit Theorem provides the theoretical foundation for statistical inference, including confidence intervals, hypothesis testing, and many approximate methods used in practice.

## Setup and Notation

Let

$$X_1, X_2, \dots, X_n$$

be a sequence of independent and identically distributed random variables such that:

$$\mathbb{E}[X_i] = \mu, \quad \text{Var}(X_i) = \sigma^2 < \infty.$$

Define the sample mean as:

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i.$$

The Central Limit Theorem concerns the distributional behavior of the normalized sample mean as the sample size increases.

# Statement of the Central Limit Theorem

The Central Limit Theorem states that:

$$\sqrt{n} \left( \frac{\bar{X}_n - \mu}{\sigma} \right) \xrightarrow{d} \mathcal{N}(0, 1) \quad \text{as } n \rightarrow \infty.$$

Equivalently, for large  $n$ ,

$$\bar{X}_n \approx \mathcal{N} \left( \mu, \frac{\sigma^2}{n} \right).$$

This result holds regardless of the original distribution of the  $X_i$ , provided the assumptions of independence, identical distribution, and finite variance are satisfied.

## Interpretation

The Central Limit Theorem implies that the distribution of the sample mean becomes approximately Normal as the sample size increases.

Key implications include:

- The mean of the distribution of  $\bar{X}_n$  is  $\mu$ .
- The variance of  $\bar{X}_n$  is  $\sigma^2/n$ .
- The dispersion of the sample mean decreases as the sample size increases.

The theorem explains why averages of random variables tend to exhibit Normal behavior even when the underlying data is skewed or discrete.

## What the CLT Does Not State

It is important to clarify several common misconceptions:

- The CLT does not state that the original data becomes Normally distributed.
- The CLT does not guarantee Normality for small sample sizes.
- The CLT does not apply to distributions with infinite variance.

The theorem applies to the distribution of the normalized sum or average, not to individual observations.

## Examples

### Bernoulli Random Variables

Let  $X_i$  be Bernoulli random variables with parameter  $p$ . Then:

$$\mu = p, \quad \sigma^2 = p(1 - p).$$

By the Central Limit Theorem,

$$\bar{X}_n \approx \mathcal{N}\left(p, \frac{p(1-p)}{n}\right) \quad \text{for large } n.$$

This explains the Normal approximation to the Binomial distribution.

### Exponential Random Variables

Let  $X_i \sim \text{Exponential}(\lambda)$ . Then:

$$\mu = \frac{1}{\lambda}, \quad \sigma^2 = \frac{1}{\lambda^2}.$$

Despite the strong skewness of the Exponential distribution, the sample mean satisfies:

$$\bar{X}_n \xrightarrow{d} \mathcal{N}\left(\frac{1}{\lambda}, \frac{1}{n\lambda^2}\right).$$

## Conditions for the Central Limit Theorem

The classical form of the Central Limit Theorem requires:

- Independence of the random variables,
- Identical distribution,
- Finite second moment.

If these conditions are violated, convergence to the Normal distribution may fail or lead to different limiting distributions.

## Central Limit Theorem vs Law of Large Numbers

The Central Limit Theorem and the Law of Large Numbers address distinct questions:

- The Law of Large Numbers concerns convergence of  $\bar{X}_n$  to  $\mu$ .
- The Central Limit Theorem concerns the distribution of fluctuations around  $\mu$ .

Both results are fundamental, but they describe different aspects of asymptotic behavior.

## Conclusion

The Central Limit Theorem explains the emergence of Normal distributions in statistical analysis and provides the theoretical foundation for much of statistical inference.

Understanding both the power and the limitations of the CLT is essential for responsible modeling and interpretation of data, particularly in applications where assumptions may be violated.